

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/348480226>

Unmanned Ground Vehicle Control using IoT

Conference Paper · December 2020

DOI: 10.1109/REM49740.2020.9313890

CITATIONS

3

READS

1,025

3 authors, including:



Tarek A. Tutunji

Al Hussein Technical University

64 PUBLICATIONS 683 CITATIONS

SEE PROFILE

Unmanned Ground Vehicle Control using IoT

This is a draft. The final paper can be found on IEEEExplore

Tarek A. Tutunji
Mechatronics Engineering Dep.
Philadelphia University
Amman, Jordan
ttutunji@philadelphia.edu.jo

Mohammed Salah-Eddin
Mechatronics Engineering Dep.
Philadelphia University
Amman, Jordan

Hisham Abdalqader
Mechatronics Engineering Dep.
Philadelphia University
Amman, Jordan

Abstract— A small unmanned ground vehicle (UGV) prototype is designed, assembled, and tested. The objective is to implement a control system for UGV via Internet of Things (IoT) to explore uncharted territory while streaming/recording videos. The UGV has an IP address and communicates with the operator through the cloud. The software architecture has two main modes: manual and autonomous. In manual mode, the operator controls the speed and direction of the vehicle's movement using an Android application called Blynk platform. The operator's control decision is based on the live-streaming. In autonomous mode, a control algorithm uses information from three ultrasonic sensors and GPS to avoid obstacles and explore the area while recording the videos in the memory. The autonomous option is used as a backup safety option in case of internet disconnection. The hardware and software design uses two controllers: Raspberry Pi and Arduino. The Raspberry Pi is programmed in Python and used for communication between the UGV/camera and the cloud. The Arduino Mega is programmed in C++ and used for the UGV's low-level Proportional-Integral-Derivative (PID) motor control and high-level autonomous control.

Keywords—UGV, IoT, Arduino, Raspberry Pi, Autonomous Systems, Cloud Computing

I. INTRODUCTION

The design and implementation of mobile robots has gained much attention in the research and industrial community. The availability of off-the-shelf robotic components, mechanical platforms, and open-source coding has increased the interest of engineers and hobbyists as many started to build their own robotic systems. Furthermore, the development of fast-processing controllers, communication protocols, and intelligent algorithms resulted in the increase of research activities in the area.

Mobile robots have a wide range of applications that include: transportation, exploration and surveillance, gaming competitions, toys, agriculture, and military. In this work, a mobile robot was developed to explore unknown area and record the environment [1].

Low cost autonomous UGV's are being developed by many researchers [2] while IoT technology is providing enhanced applications for mechatronic systems, such as robotics [3,4]. Furthermore, new algorithms, such as deep learning and multi-rate control strategies, are being developed for mobile robotic research work [5, 6].

In this work, the developed mobile robot system is interfaced with the user via the internet as the user-controller signals and video recording is exchanged through the cloud. The system also has an autonomous option where the robot

uses an embedded algorithm to explore the area, record the video on its memory, and find its way to the base.

The paper is formulated as follows: Section Two identifies the system specifications; Section Three provides the hardware architecture design; Section Four provides the software design; Section Five discusses the prototype testing; Section Six concludes the paper and provides future recommendations.

II. SYSTEM SPECIFICATIONS

The main system requirements were set as follows: Manual control using mobile phone application and communicating via the internet; Autonomous control to avoid obstacles, record video, and return to home position.

As an initial step, a small-sized UGV prototype specifications were set as shown in Table 1. Once the algorithm is developed and tested, a CAD-based mechanical system will be designed and built.

Table 1. Prototype Specifications

Dimension	25x14x10 cm
Overall Weight	2 Kg
Maximum Velocity	0.5 m/s
Camera resolution	5 Mega-pixel
Compass accuracy	1 to 5 degress
Position accuracy	+ - 2 – 4 m
Main power	12VDC lithium batteries
Code language	C++ and Python
Cost	300 \$

The electronic system will have motors to drive the UGV and another motor to move the camera. Sensors will be used to detect obstacles and identify the UGV's location and direction of movement. An android-based application will be used to interface with the UGV via the cloud by exchanging the video feed control signals. The general block diagram is shown in Fig. 1

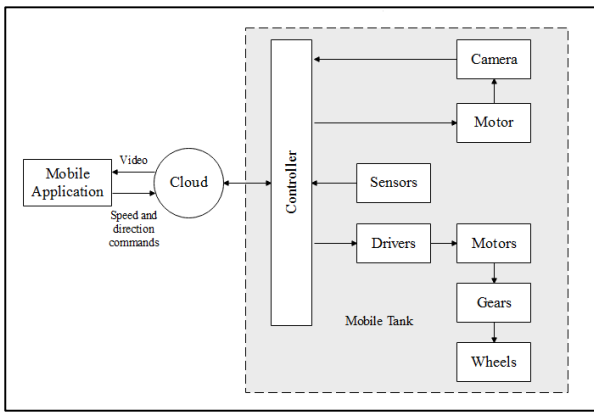


Fig. 1 General Block Diagram

III. HARDWARE DESIGN

In order to implement the system requirements, two microcontroller platforms were used: Raspberry Pi and Arduino.

The Raspberry main tasks are: Check for the internet connection; Receive data from Android application and send it to the Arduino to drive the UGV; Live streaming in the manual mode and recording the video in the autonomous mode.

The Arduino main tasks are: Receive the command data from the Raspberry Pi to make action as for control mobile robot direction or control camera direction in the manual mode; Check battery charge capacity; Save starting point coordinates; Mathematical calculation between two coordinate point; Obstacle avoidance in the autonomous mode.

The detailed block diagram is shown in Fig. 2

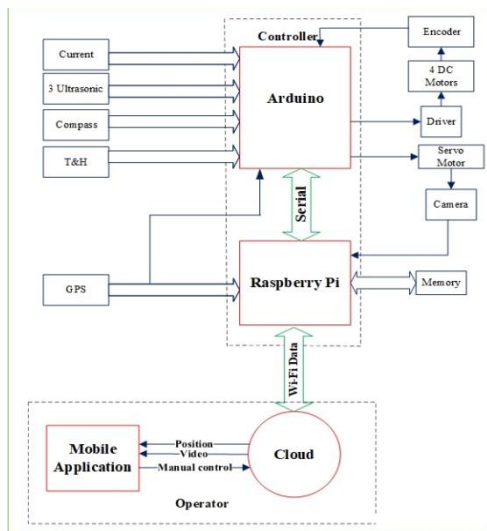


Fig. 2 System Block Diagram

A Raspberry camera was used to record the video; A compass and GPS circuit were used to identify the location and direction of the vehicle; Three ultrasonic sensors to detect obstacles; Four DC motors with two dual drive circuits were used to control the UGV speed and direction; A servo motor to control the angle of the camera. The schematic diagram and component interface is shown in Fig. 3 while the Arduino connections are shown in Fig. 4

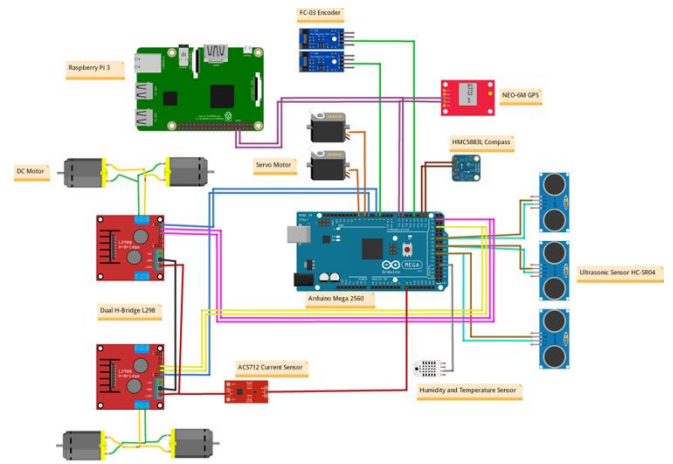


Fig 3. System Interface and connections

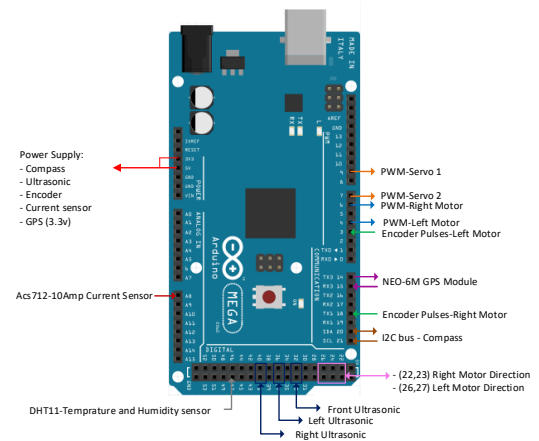


Fig 4. Arduino pin connections

IV. SOFTWARE DESIGN

The software design was divided into two main parts: The Raspberry Pi was the main processor as it selects between the manual and autonomous modes. It communicates between the Arduino and the Android Blynk application through the cloud. It sends the user's control signal to the Arduino and transmits the recorded video to the cloud. Furthermore, it controls the camera's angle and communicates with the GPS information.

On the other hand, the Arduino controls the motors' speeds and therefore, the vehicle's speed and direction. It also receives and processes the ultrasonic signals. The software structure is shown in Fig. 5

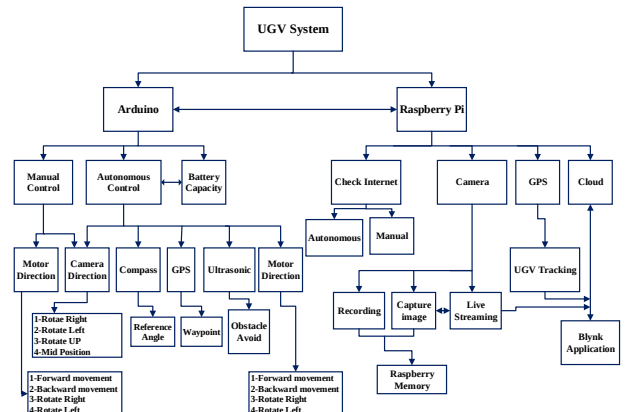


Fig 5. Software Structure

The software was divided into several parts as the main program calls several subroutines. These include *Autonomous* and *Manual* subroutines for the Arduino that in turn uses *Manual Drive* and *Camera Control* subroutines. Also, there are *Autonomous* and *Manual* subroutines for the Raspberry that includes *Stream Live Video*, *UGV Tracking*, and *Direction Command* subroutines. The system's flowchart is shown in Fig 6.

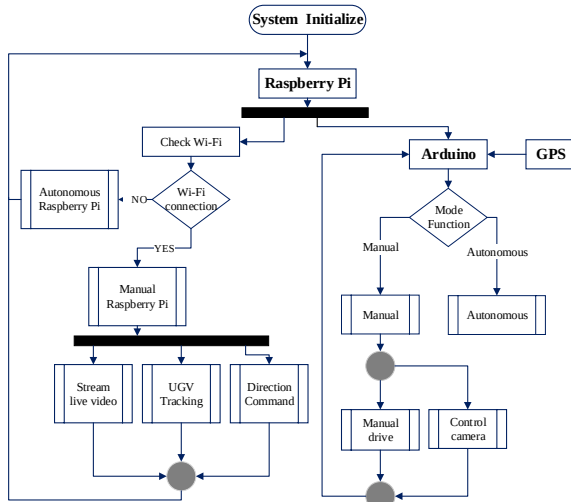


Fig. 6 Main Flowchart

The UGV's movement was driven by four motors which were controlled using PID control method. The logic control was based on the ultrasonic signals as shown in Table 2.

Table 2. Motor Logic Control

Sensor Feedback Signals			Robot Command				
FR	FM	FL	Action	Motor R1	Motor R2	Motor L1	Motor L2
0	0	0	Forward	1	1	1	1
X	1	X	Backward	-1	-1	-1	-1
0	X	1	Right	-1	-1	1	1
1	X	0	Left	1	1	-1	-1

Bug Algorithms have been successfully used in mobile robotics to avoid obstacles while tracking their target. The main idea is for the robot to move toward the goal directly according to closest line between position and goal. When faced with an obstacle, the robot adjusts its path accordingly [1]. A Dist-Bug algorithm was implemented, where Dist-Bug theory consist of, when the robot encounters an obstacle in its path, it starts following the edge of the obstacle simultaneously calculating and storing the distance from its current and next position to destination. This is shown in Fig. 7

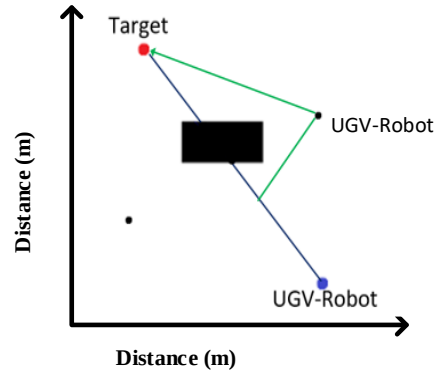


Fig. 7 Dist-Bug Illustration

V. PROTOTYPE

Two prototypes were assembled: Prototype One was used for manual control only through the IoT interface while Prototype Two was used for manual and autonomous control. The two prototypes were assembled by different student teams. Table 3 shows the comparison between the prototypes.

For Prototype One, Python and HTML programming languages were used to implement the code. The serial communication between the Raspberry Pi and the Arduino used the Numpy library as it can control all the Arduino pin from the Raspberry Pi in the Python script. Arduino worked as a slave controlled by a master Raspberry Pi, where Blynk ,an Android application, was used to facilitate the communication between the Raspberry and the Arduino as it allowed the user to remotely control the UGV over the internet. Fig. 8 shows the prototype system while Fig. 9 shows the system interface (Blynk) [7].

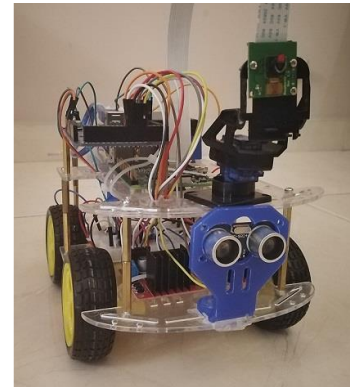


Fig. 8. Prototype One

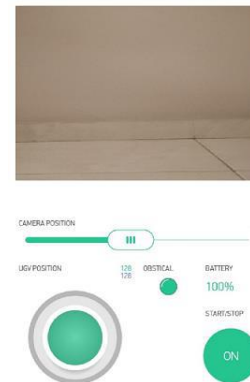


Fig. 9. Prototype One User Interface

For Prototype Two, C++, Python and HTML programming languages were used to implement the code. The Raspberry Pi was programed with Python and HTML while the Arduino was programed with using C++ code. Therefore, the Nanpy library was no longer used. In manual mode, the Raspberry Pi was the master and Arduino was the slave. In autonomous mode, the Arduino worked as the master for making decisions such as anti-collision and for scanning area or go back to home position according to the battery charge capacity. The interface Android application (Blynk) was modified to fit the system requirement such as broadcasting, UGV tracking and remotely control mobile robot shown in Fig. 10 shows the built prototype while Fig. 11 shows the modified user interface [8].

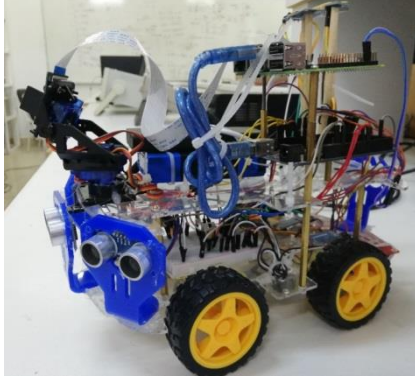


Fig. 10 Prototype Two

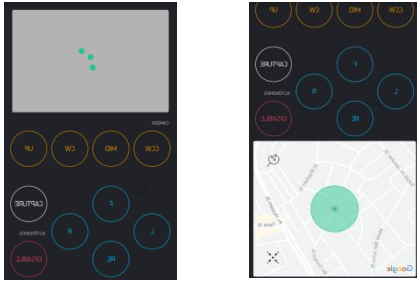


Fig. 11 User Interface

Comparisons between the prototypes is summarized in Table 3.

Table 3. Comparison between Prototypes

	Prototype One	Prototype Two
Modes	Manual	Manual and Autonomous
Hardware Mechanical	Four wheels driven by four geared DC motors.	Four wheels driven by four geared DC motors.
Hardware Electrical	Two Dual H-Bridge drivers.	Two Dual H-Bridge drivers.
Microcontrollers	Raspberry Pi 3 (Model B).	Raspberry Pi 3 (Model B).
	Arduino Mega.	Arduino Mega.
Camera	Raspberry Pi Camera.	Raspberry Pi Camera.
	Movable camera mount with servo motor	Movable camera mount with servo motor
Sensors	Two ultrasonic sensors.	Three ultrasonic sensors.
	Current sensor.	Current sensor.
	---	Compass
	---	GPS module
	---	Temperature and Humidity sensor
Battery	12v Batteries	12v Batteries
Software	Python	Python and C++
Android Application	Blynk user interface One	Blynk user interface Two

An example of the vehicle path control movement was captured in Fig. 12

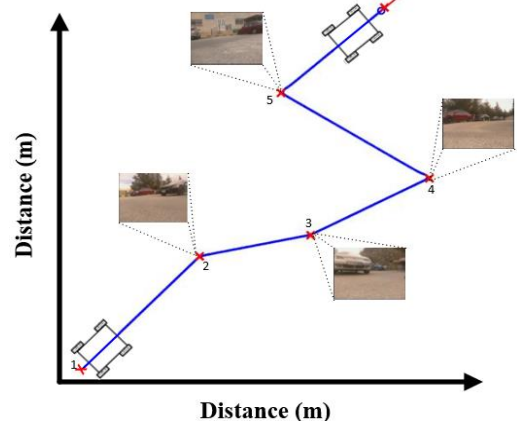


Fig. 12 Mobile Robot Path

The low-level control for both prototypes was implemented using PID control. The set speed reference for the motors was 180 RPM. Due to the lack of the necessary data about the mobile-robot motor, the PID parameter were chosen manually using trial and error test. These parameters were set to ($K_p = 0.10$, $K_d = 0.005$, $K_i = 0.005$). Fig. 13 shows the system response as the desired speed was set to 180 RPM.

To observe appropriate system behavior, the system was tested under different load forces that were applied on the motor wheel from time (150 to 430 s). The results are shown in Fig. 14.

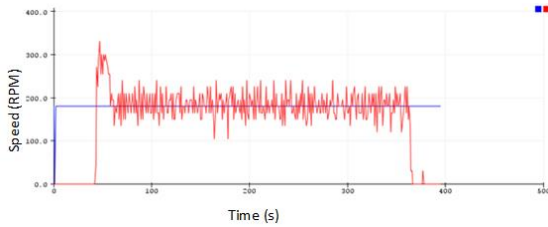


Fig. 13 Closed-Loop Speed Control without Disturbance

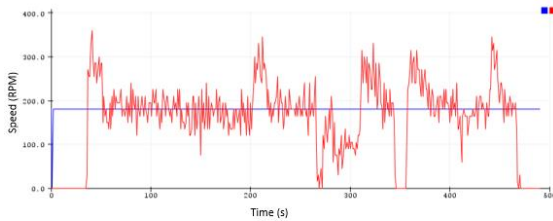


Fig. 14 Closed-Loop Speed Control with Disturbance

The proposed communication and control algorithm was implemented on mini-mobile robot for indoor application as a proof of concept.

VI. CONCLUSION AND FUTURE WORK

A UGV prototype control algorithm was developed, implemented, and tested successfully. Two prototypes were built and tested for manual and autonomous modes. In the manual mode, the Raspberry Pi was used for the main communication and the high-level control while the Android for the low-level control. In the autonomous mode, the Raspberry Pi was used to record the video while the Android was used to make decisions (i.e. high-level) and drive the UGV's motors (i.e. low-level control).

Future work will have two main parts: Design a rigid medium-sized CAD-based mechanical robot structure and develop new algorithms. Research is underway to develop and implement intelligent algorithms, vehicle-to-vehicle communication, and optimize the coding using object-oriented tools.

ACKNOWLEDGMENT

This work was funded by Philadelphia University. Seven students were involved in the design and implementation of the UGV. The project was divided into three parts: electrical, mechanical, and software control. This paper concentrates on the software part only.

REFERENCES

- [1] R. Siegwart, I. Nourbakhsh and D. Scaramuzza, , "Introduction to Autonomous Mobile Robots" 2nd edition, MIT Press 2011
- [2] C. Man, Y. Koonjul, and L. Nagowah "A low cost autonomous unmanned ground vehicle", Future Computing and Informatics Journal 3 (2018) 304-320
- [3] D. Bradley, D. Russel, I. Ferguson, J. Isaacs, A. MacLead, and R. White, "The Internet of Things – The future or the end of mechatronics", Mechatronics 27 (2015) 57-74
- [4] L.A. Greico, A. Rizzo, S. Colluci, G. Sicari, G. Piro, D. Di Paolo, and G. Boggia, "IoT-aided robotics applications: Technological implications, target domains and open issues" Computer Communications 54 (2014) 32-47..
- [5] N. Patel, A. Choromanski, P. Krishnamurthy, and F. Khorrami, "A deep learning gated architecture for UGV navigation robust to sensor", Robotics and Autonomous Systems 116 (2019) 80-97
- [6] J. Salt, J. Alcaina, A. Cuenca, and A. Barrios, "Multirate control strategies for avoiding sample losses. Application to UGV path tracking" ISA Transactions, Article in Press
- [7] H. Abdalqadar, M. Abo Siam, H. Altalla, and A. Harbi, "Mobile Tank Vehicle Control using IoT", Senior Student Project, Philadelphia University, 2019
- [8] M. Salah-Eddin and A. Abushaar, "Unmanned Ground Vehicle Control using IOT: Phase-Two. Senior Student Project, Philadelphia University, 2020

